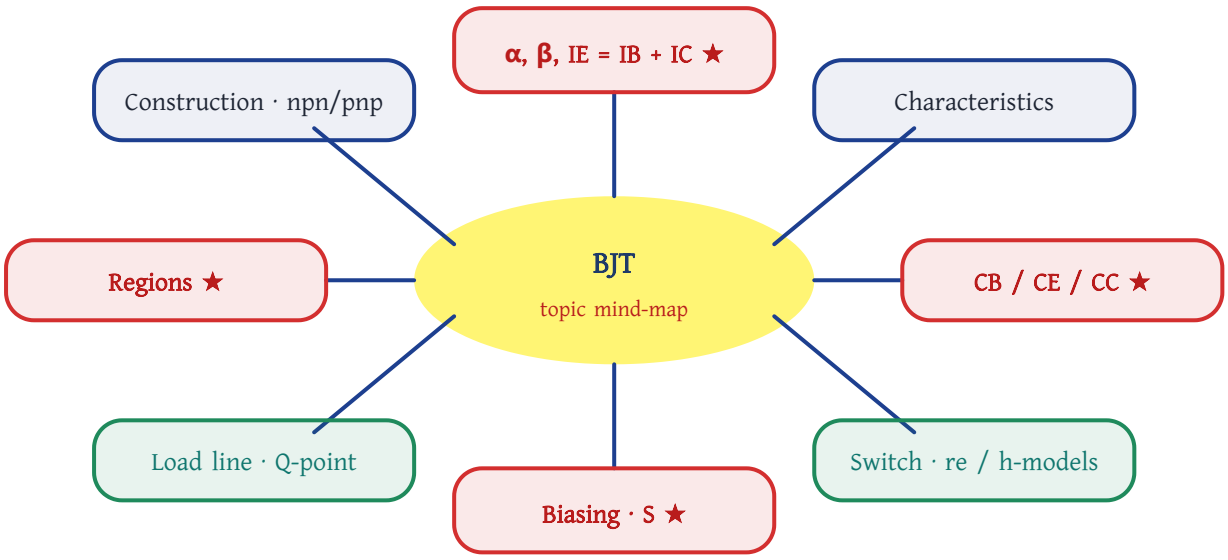
Bipolar Junction Transistor

Construction · α, β · regions · CB/CE/CC · load line · biasing · switch · models: exam-ready in 11 sheets



↑ the whole topic on one page: red = high-yield, green = method / applied

Syllabus coverage

"**Bipolar junction transistors**: construction and operation, amplifying action, common base, common emitter, common collector, **load line**, different biasing, **stability factor**, small signal equivalent circuit models, BJT as a switch". Gain, input and output impedance of the single-stage amplifiers are worked out in the Single Stage Amplifiers & FET notes; diodes are in Diode Circuits.

High-yield list

① doping E > C > B
② $\beta = \alpha/(1 - \alpha)$
③ $I_{CEO} = (1 + \beta)I_{CBO}$
④ junction bias per region
⑤ CB/CE/CC table
⑥ S: fixed = 1 + β
⑦ re = 26 mV / IE

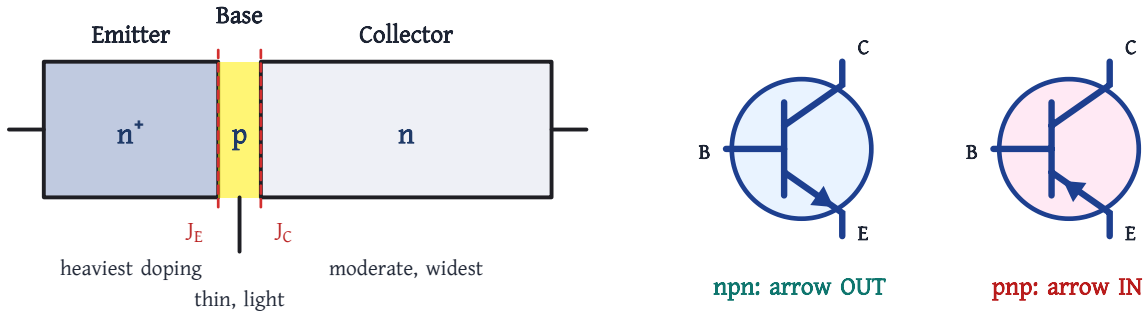
Legend: **yellow** = key term · **red circle** = must-memorise fact · **purple box** = memory trick · **red box** = exam trap · **green** = worked example

Contents

✓ Construction, npn / pnp ..... p. 2
✓ Operation, α, β, leakage ..... p. 3
✓ Regions & characteristics ..... p. 4
✓ CB, CE, CC configurations ..... p. 5
✓ DC load line & Q-point ..... p. 6
✓ Biasing & stability factor ..... p. 7
✓ Bias examples, thermal runaway ... p. 8
✓ Switch & small-signal models ... p. 9
✓ Night-before sheet ..... p. 10
✓ Practice MCQs + written ..... p. 11

1 Construction: Three Layers, Two Junctions

Definition: A BJT is three doped regions, **emitter (E), base (B), collector (C)**, forming two p–n junctions: emitter–base (J_E) and collector–base (J_C). **Bipolar** = both electrons and holes carry the current (a FET is unipolar). It is a **current-controlled** device: a small I_B controls a large I_C.



npn layer sandwich (not to scale: the base is far thinner) and the two symbols

Region	Doping	Width	Job
Emitter	**heaviest**	medium	injects (emits) majority carriers into the base
Base	lightest	**thinnest** (μm)	passes carriers through with little recombination
Collector	moderate	**widest**	collects carriers, dissipates most of the heat

Order to memorise: doping **E > C > B** · width **C > E > B**. A thin, lightly doped base means few recombinations, so $I_C \approx I_E$.

	npn	pnp
Majority carrier in E	electrons	holes
Normal supply	C +, $V_{BE} \approx +0.7$ V	C −, $V_{EB} \approx +0.7$ V
Emitter arrow	out (conventional I leaves E)	in
Speed	**faster** ($\mu_n \approx 2.8\ \mu_p$ in Si)	slower

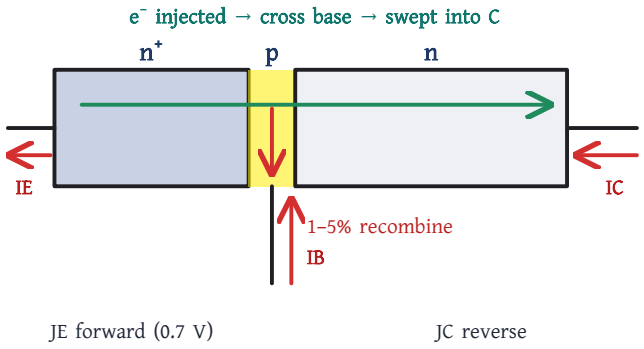
Memory trick: npn = "**N**ot **P**ointing i**N**"; pnp = "**P**oints i**N** **P**roudly". The arrow is always on the **emitter** and points p → n.

Exam trap: E and C of an npn are both n-type but **not interchangeable** (doping and size differ). Two separate diodes back to back do **not** make a transistor: the base must be thin.

History: point-contact transistor, Bardeen & Brattain, Bell Labs, **1947**; junction (bipolar) transistor, Shockley, 1948; Nobel Prize 1956. "Transistor" = **transfer + resistor**. Silicon replaced germanium: lower leakage, works to ≈ 150 °C.

2 Operation (npn, Active Mode)

1. J_E **forward**-biased: the n^+ emitter injects electrons into the base → I_E
2. The base is thin and lightly doped: only **1–5%** recombine with holes → I_B
3. J_C **reverse**-biased: its field sweeps the other 95–99% into the collector → I_C
4. A small I_B controls a large I_C: this is **amplifying action**



Current relations: $I_E = I_B + I_C$ · $\alpha = I_C/I_E$ (0.95–0.995) · $\beta = I_C/I_B$ (50–300)
$\beta = \alpha/(1 - \alpha)$ · $\alpha = \beta/(1 + \beta)$ · $I_E = (1 + \beta)I_B$ · $\gamma = I_E/I_B = 1 + \beta$ (CC)

Leakage currents

- I_{CBO}: C–B reverse current with the **emitter open** (Si: nA, Ge: μA); minority carriers, **doubles every 10 °C**
- CB: $I_C = \alpha I_E + I_{CBO}$
- CE: $I_C = \beta I_B + (1 + \beta)I_{CBO} = \beta I_B + I_{CEO}$
- I_{CEO} (base open) = $(1 + \beta)I_{CBO}$: β times worse

dc vs ac β: $\beta_{dc} = h_{FE} = I_C/I_B$ · $\beta_{ac} = h_{fe} = \Delta I_C/\Delta I_B$; nearly equal in the active region.

Memory trick: **α** = "Almost 1", **β** = "Big". β = "a over one-minus-a".

Worked examples 1–2 ✍️: I_E = 10 mA, I_C = 9.9 mA → I_B = 0.1 mA, α = **0.99**, β = **99** (0.99/0.01 = 99 ✓) · α = 0.98 → β = 0.98/0.02 = **49** · β = 200 → α = 200/201 = **0.995**

Worked example 3 ✍️: Ge transistor, β = 50, I_{CBO} = 5 μA, I_B = 40 μA → I_{CEO} = 51 × 5 = **255 μA**; I_C = 50 × 40 + 255 = 2255 μA ≈ **2.26 mA**. Leakage is 11% of I_C, which is why Ge circuits drift with heat.

Worked example 4 ✍️: Si, β = 120, I_B = 50 μA → I_C = **6 mA**, I_E = 121 × 50 μA = **6.05 mA**, α = 120/121 = 0.9917, so $I_C \approx I_E$.

Exam trap: β is **not** a constant: it rises with temperature, varies with I_C, and spreads widely between units of one type (e.g. h_{FE} 100–300 on one datasheet). Good bias design must not depend on it.

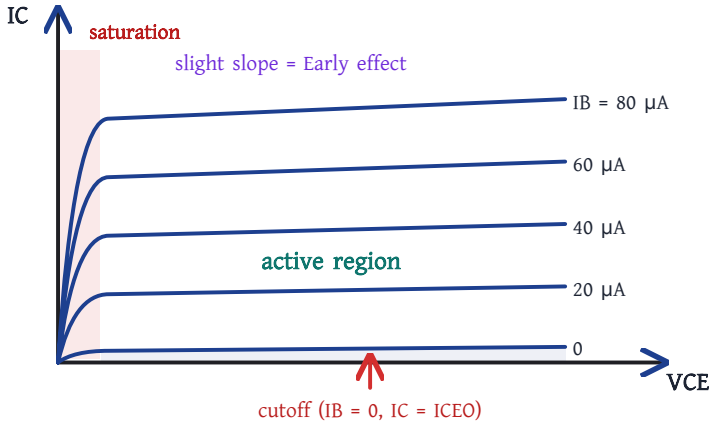
3 Regions of Operation

Region	J_E (B–E)	J_C (B–C)	npn values	Use
Cutoff	reverse	reverse	$V_{BE} < 0.5$ V, $I_C \approx I_{CEO} \approx 0$	switch OFF
Active	**forward**	**reverse**	$V_{BE} \approx 0.7$ V, $I_C = \beta I_B$	amplifier
Saturation	forward	forward	$V_{CE(sat)} \approx 0.2$ V, $I_C < \beta I_B$	switch ON
Inverse active	reverse	forward	β very small (≈ 1)	rarely (TTL input)

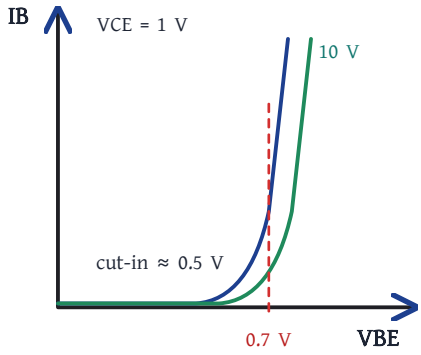
Memory trick: Read (J_E, J_C): **F-R** = "**F**or **R**eal gain" (active) · **F-F** = fully ON · **R-R** = really OFF.

Exam trap: In saturation $I_C = \beta I_B$ is **false**: I_C is set by the circuit ($\approx V_{CC}/R_C$) and is smaller than βI_B.

4 CE Characteristics



output: IC vs VCE for steps of IB



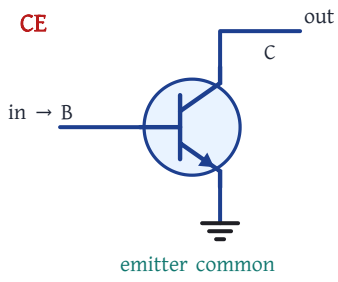
input: a forward diode curve

- **Early effect** (base-width modulation): V_{CE} ↑ → C–B depletion widens → effective base narrower → β and I_C rise a little. Extended curves meet the V_{CE} axis at $-V_A$ (Early voltage, 50–200 V); output resistance $\mathbf{r_o = V_A/I_C}$ (V_A = 100 V, I_C = 1 mA → 100 kΩ).
- The same effect shifts the input curve **right** as V_{CE} rises (less I_B for the same V_{BE}).
- **CB characteristics:** input I_E vs V_{BE} (diode); output $I_C \approx I_E$, almost flat, and I_C still flows at $V_{CB} = 0$ (needs a small forward V_{CB} to stop).
- **Breakdown:** $BV_{CEO} < BV_{CBO}$; **punch-through** when the C–B depletion layer reaches the emitter.

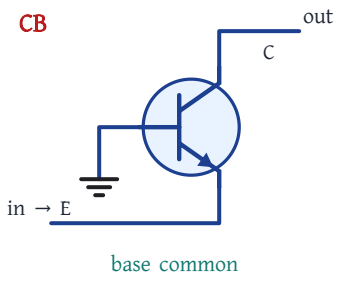
Worked example: active or saturated? ✍️: V_{CC} = 10 V, R_C = 1 kΩ, β = 100, I_B = 0.2 mA. βI_B = 20 mA, but $I_{C(sat)} = (10 - 0.2)/1$ kΩ = 9.8 mA < 20 mA → **saturated**, $V_{CE} \approx 0.2$ V, I_C = 9.8 mA.

5 Three Configurations: CB, CE, CC

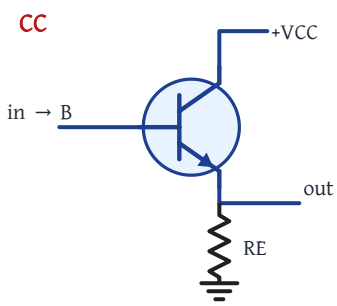
The name tells which terminal is **common** to input and output (tied to AC ground).



emitter common



base common



	CB	CE	CC (emitter follower)
Input → output	E → C	B → C	B → E
Current gain	α (< 1, ≈ 0.99)	**β (high)**	1 + β (highest)
Voltage gain	high	**high**	≈ 1 (slightly less)
Power gain	medium	**highest**	low
Input impedance	very low (≈ 20–50 Ω)	medium (≈ 1 kΩ)	**very high (≈ βR_E)**
Output impedance	**very high (≈ MΩ)**	medium-high (≈ 10–50 kΩ)	very low (≈ tens of Ω)
Phase (in → out)	0°	**180°**	0°
Leakage	I_{CBO}	I_{CEO}	I_{CEO}
Main use	RF/VHF, cascode, current buffer	general voltage amplifier	buffer, impedance matching

Memory trick: CB = current gain **B**elow 1 · CE = gains **E**verything, and inverts · CC = **C**opies the voltage (follower).

Exam trap: CC current gain is **1 + β**, not β; its voltage gain is **just under 1**, not exactly 1. CB has a **high** voltage gain despite α < 1.

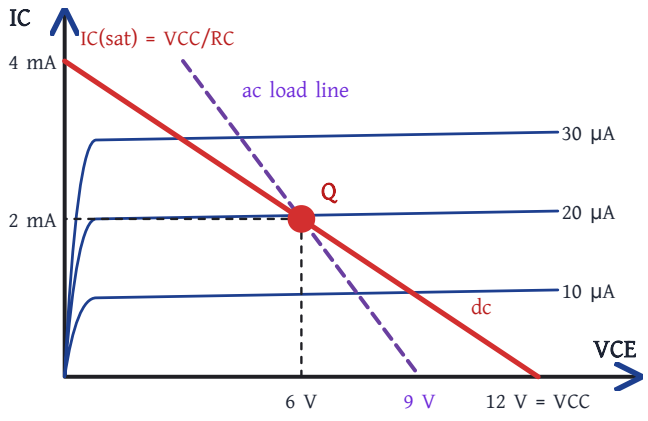
Amplifying action (CB): voltage gain with no current gain

Worked example ✍️: v_i = 20 mV across r_i = 20 Ω → I_e = 1 mA; $I_c = \alpha I_e \approx 1$ mA flows in R_L = 5 kΩ → v_L = 5 V. A_v = 5 V/20 mV = **250**, $A_i \approx 1$, $A_p = A_v A_i \approx 250$. The current is **transferred** from a low-resistance input to a high-resistance output: "transfer resistor".

Darlington pair (two CC stages): $\beta_{total} \approx \beta_1\beta_2$ (100 × 100 = 10 000), $V_{BE} \approx 1.4$ V. Gains, Z_{in}, Z_{out} with h-parameters: see Single Stage Amplifiers & FET.

6 DC Load Line & Q-point

Definition: The **DC load line** is the set of all (V_{CE}, I_C) the external circuit allows: $\mathbf{V_{CE} = V_{CC} - I_C(R_C + R_E)}$ (R_E = 0 if there is none). The **Q-point** (quiescent point) is where it meets the transistor's I_B curve with no signal applied.



dc load line (red), ac load line (purple)

- **Saturation end** (V_{CE} = 0): $I_{C(sat)} = V_{CC}/(R_C + R_E)$
- **Cutoff end** (I_C = 0): $V_{CE(off)} = V_{CC}$
- Slope = $-1/(R_C + R_E)$
- Q near saturation → output clipped at the bottom ($V_{CE(sat)}$); near cutoff → clipped at the top (V_{CC})
- Q in the **middle** → largest symmetrical swing (class A)

Worked example (fixed bias) ✍️: V_{CC} = 12 V, R_B = 565 kΩ, R_C = 3 kΩ, β = 100, V_{BE} = 0.7 V.

1. $I_B = (12 - 0.7)/565\ k\Omega =$ **20 µA**; $I_C = \beta I_B =$ **2 mA**
2. $V_{CE} = 12 - 2\ mA \times 3\ k\Omega =$ **6 V** → Q = (6 V, 2 mA)
3. Ends: $I_{C(sat)} = 12/3\ k\Omega = 4$ mA, $V_{CE(off)} = 12$ V → Q exactly mid-line

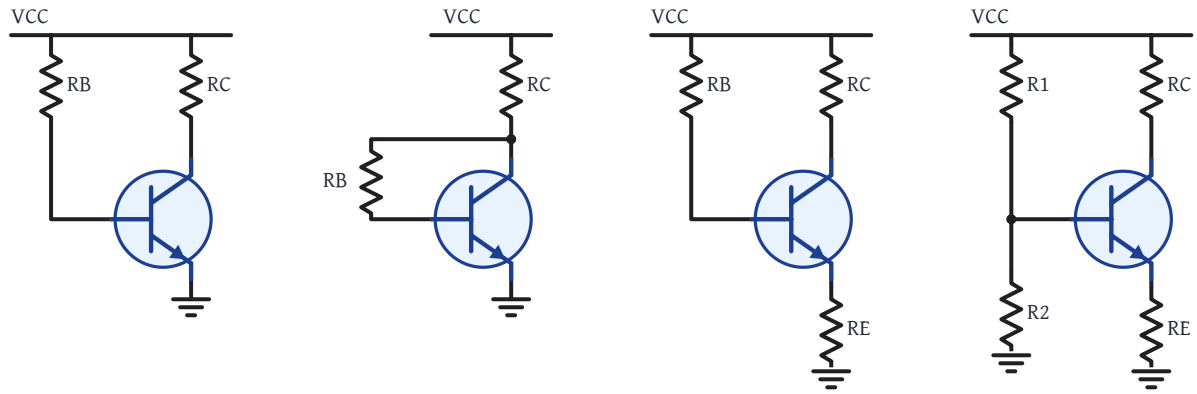
Change	Effect on line / Q
R_B ↑	I_B ↓: Q slides down the same line toward cutoff
R_C ↑	line flatter ($I_{C(sat)}$ ↓); Q moves toward saturation
V_{CC} ↑	line shifts outward, parallel (same slope)
β ↑ (heat, new transistor)	fixed bias: I_C ↑, Q climbs toward saturation

AC load line ✍️: With a load R_L = 3 kΩ coupled by a capacitor, $r_{ac} = R_C \parallel R_L$ = 1.5 kΩ; the steeper ac line passes through Q: $i_{c(sat)} = I_{CQ} + V_{CEQ}/r_{ac} = 2 + 6/1.5 = 6$ mA, $v_{ce(off)} = V_{CEQ} + I_{CQ}r_{ac} = 6 + 3 = 9$ V. Largest undistorted peak = min(6, 3) = **3 V**.

Exam trap: Same circuit, β = 200: $\beta I_B = 4$ mA would need $V_{CE} = 0$, so it **saturates** ($I_C \approx 11.8\ V/3\ k\Omega = 3.93$ mA). A fixed-bias Q-point depends entirely on β.

7 Biasing Methods & Stability Factor

Why bias?: To place Q in the active region and **keep it there** while β (unit spread, temperature), **I_{CBO} (doubles / 10 °C)** and V_{BE} (falls ≈ 2.5 mV/°C) change.



① fixed ② collector-feedback ③ emitter-stabilised ④ voltage-divider

Stability factors: $S = \partial I_C/\partial I_{CBO}$ (β, V_{BE} constant) · $S' = \partial I_C/\partial V_{BE}$ · $S'' = \partial I_C/\partial \beta$. Range $\mathbf{1 \le S \le 1 + \beta}$; **smaller S = more stable**. General: $S = (1+\beta)/(1 - \beta\cdot\partial I_B/\partial I_C)$

Method	I_B	V_{CE}	S
① Fixed	$(V_{CC} - V_{BE})/R_B$	$V_{CC} - I_C R_C$	$1 + \beta$ (worst)
② Coll.-feedback	$(V_{CC} - V_{BE})/(R_B + (1+\beta)R_C)$	$V_{CC} - I_E R_C$	$(1+\beta)/(1+\beta R_C/(R_C + R_B))$
③ Emitter-stab.	$(V_{CC} - V_{BE})/(R_B + (1+\beta)R_E)$	$\approx V_{CC} - I_C(R_C + R_E)$	$(1+\beta)(1+R_B/R_E)/(1+\beta+R_B/R_E)$
④ Volt.-divider	$(V_{Th} - V_{BE})/(R_{Th} + (1+\beta)R_E)$	$\approx V_{CC} - I_C(R_C + R_E)$	same, $R_B \to R_{Th}$; $\approx 1 + R_{Th}/R_E$

④: $R_{Th} = R_1 \parallel R_2$, $V_{Th} = V_{CC}R_2/(R_1 + R_2)$. **Approximate method** when $\beta R_E \ge 10R_2$: $V_B = V_{Th}$, $I_E = (V_B - 0.7)/R_E$, independent of β. Most stable, most widely used.

Derivation ✍️: ③: $V_{CC} = I_B R_B + V_{BE} + (I_B + I_C)R_E \to \partial I_B/\partial I_C = -R_E/(R_B + R_E) \to S = (1+\beta)/(1+\beta R_E/(R_B + R_E))$. Fixed bias: I_B does not depend on I_C, so $\partial I_B/\partial I_C = 0$ and $S = 1 + \beta$.

Memory trick: R_E looks **(1+β) times bigger** from the base. Small R_B/R_E (or R_{Th}/R_E) → S → 1.

Exam trap: A bypass capacitor across R_E removes the **AC** feedback (gain comes back) but keeps the **DC** stability. Collector-feedback also cuts the AC gain.

8 Bias Worked Examples

Example A: voltage-divider ✍️: V_{CC} = 18 V, R_1 = 39 kΩ, R_2 = 8.2 kΩ, R_C = 3.3 kΩ, R_E = 1 kΩ, β = 120.

1. Check: βR_E = 120 kΩ ≥ $10R_2$ = 82 kΩ ✓ → approximate method allowed
2. V_B = 18 × 8.2/47.2 = 3.13 V; V_E = 2.43 V; $I_C \approx I_E$ = **2.43 mA**
3. V_{CE} = 18 − 2.43 × (3.3 + 1) = **7.56 V**
4. Exact: R_{Th} = 6.78 kΩ, I_B = 2.43/(6.78 + 121) = 19.0 μA, I_C = 2.28 mA, V_{CE} ≈ 8.18 V (approx. is 6% high)
5. S = 121(1 + 6.78)/(121 + 6.78) ≈ **7.4** (fixed bias would be 121)

Example B: emitter-stabilised ✍️: V_{CC} = 20 V, R_B = 430 kΩ, R_C = 2 kΩ, R_E = 1 kΩ, β = 50.

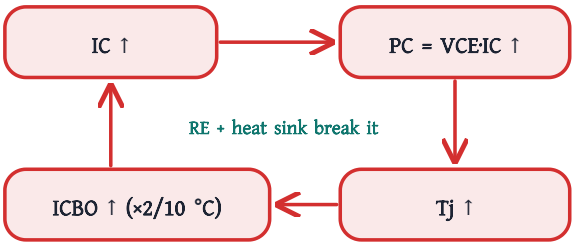
1. I_B = 19.3 V/(430 + 51 × 1) kΩ = **40.1 μA**; I_C ≈ **2.01 mA**; V_{CE} ≈ 20 − 2.01 × 3 = **13.97 V**
2. S = 51(1 + 430)/(51 + 430) ≈ 45.7: better than fixed (51), far worse than Example A

β doubles (circuit)	I_C before → after	Change
Fixed bias (any values)	$I_C \rightarrow 2I_C$	+100%
Emitter-stabilised (Ex. B, 50 → 100)	2.01 → 3.63 mA	+81%
Collector-feedback (12 V, R_B 200 k, R_C 2 k; 100 → 200)	2.81 → 3.75 mA	+34%
Voltage-divider (Ex. A, 120 → 240)	2.28 → 2.35 mA	**+3%**

9 Thermal Runaway

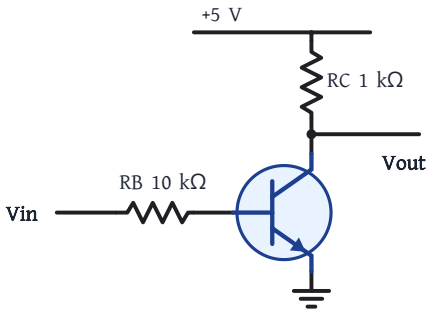
Definition: A **self-destroying positive feedback** loop: heat raises I_{CBO} (and β) → I_C rises → collector dissipation $P_C = V_{CE}I_C$ rises → the junction heats further, until the transistor burns out.

- Prevent: R_E / low S, **heat sink** (low thermal resistance θ), diode or thermistor compensation in the bias network
- No runaway if $V_{CE} < V_{CC}/2$: $P_C = I_C(V_{CC} - I_C R) \rightarrow dP_C/dI_C = 2V_{CE} - V_{CC} < 0$, so a rise in I_C **lowers** P_C ($R = R_C + R_E$)



the runaway loop

10 BJT as a Switch



Vout = (Vin)′: NOT gate

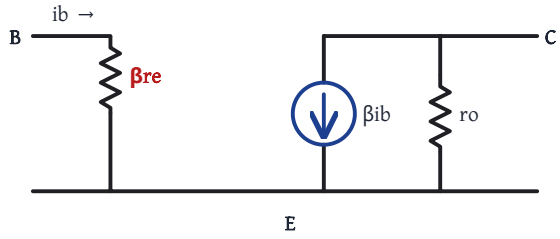
V_{in}	Region	Switch	V_{out}
0 V (0)	cutoff	open	5 V (1)
5 V (1)	saturation	closed	0.2 V (0)

Worked example ✍️: $I_{C(sat)}$ = (5 − 0.2)/1 kΩ = 4.8 mA; with β = 100, $I_{B(min)}$ = 4.8 mA/100 = 48 µA. Actual I_B = (5 − 0.7)/10 kΩ = 0.43 mA ≈ 9 × $I_{B(min)}$ → **hard saturation** ✓. R_B must stay below 4.3 V/48 µA ≈ 89.6 kΩ. Rule: **$I_B > I_{C(sat)}/\beta$**, overdrive 2–10×.

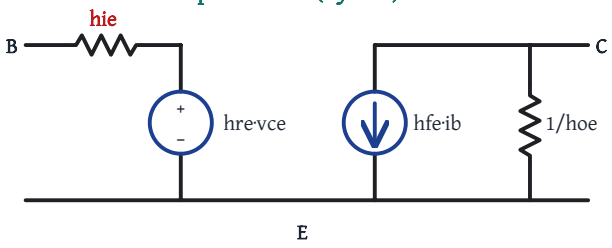
- Switching times: $t_{on} = t_d$ (delay) + t_r (rise); $t_{off} =$ **t_s (storage)** + t_f (fall). t_s is the longest: stored base charge must leave. A Schottky clamp (74LS TTL) stops deep saturation.
- Dissipation is low in both states ($I_C \approx 0$, or $V_{CE} \approx 0.2$ V); it peaks during transitions. Relay coil: add a **flyback diode**.

11 Small-Signal Models

CE re model



CE h-parameter (hybrid) model



re model: $r_e = 26\ mV/I_E$ (V_T = kT/q ≈ 26 mV at 300 K). CE input = βr_e; CB input = r_e. Hybrid-π: $g_m = I_C/V_T$, $r_\pi = \beta/g_m$.

h-parameters: $v_1 = h_i i_1 + h_r v_2$; $i_2 = h_f i_1 + h_o v_2$. Units: h_i Ω, h_r and h_f none, h_o S. "Hybrid" = mixed units.

CE	Typical	Link
h_{ie}	1–3 kΩ	= βr_e
h_{re}	≈ 10^{-4}	≈ 0 (approx.)
h_{fe}	50–300	= β; h_{fb} = −α
h_{oe}	≈ 20 µS	= $1/r_o$

Worked example ✍️: $I_C \approx I_E$ = 2 mA, β = 120, V_A = 100 V: r_e = 26/2 = **13 Ω**; $h_{ie} = r_\pi$ = 120 × 13 = **1.56 kΩ**; g_m = 2/26 = 76.9 mS; r_o = 100/2 = 50 kΩ → h_{oe} = 20 µS. Gain analysis: Single Stage Amplifiers notes.

Night-Before-the-Exam Sheet

Formula sheet:

$I_E = I_B + I_C$ · $I_E = (1 + \beta)I_B$

$\alpha = I_C/I_E$ · $\beta = I_C/I_B$

$\beta = \alpha/(1 - \alpha)$ · $\alpha = \beta/(1 + \beta)$

$I_{CEO} = (1 + \beta)I_{CBO}$

$I_C = \beta I_B + (1 + \beta)I_{CBO}$

I_{CBO} doubles every 10 °C

V_{BE} 0.7 V (Si), 0.3 V (Ge)

$V_{CE(sat)} \approx 0.2$ V

Load line: $V_{CE} = V_{CC} - I_C(R_C + R_E)$

$I_{C(sat)} = V_{CC}/(R_C + R_E)$

$S = \partial I_C/\partial I_{CBO}$; fixed: $1 + \beta$

VD: $S \approx 1 + R_{Th}/R_E$; $\beta R_E \geq 10R_2$

$r_e = 26\ mV/I_E$ · $h_{ie} = \beta r_e$

$g_m = I_C/V_T$ · $r_o = V_A/I_C$

Switch: $I_B > I_{C(sat)}/\beta$

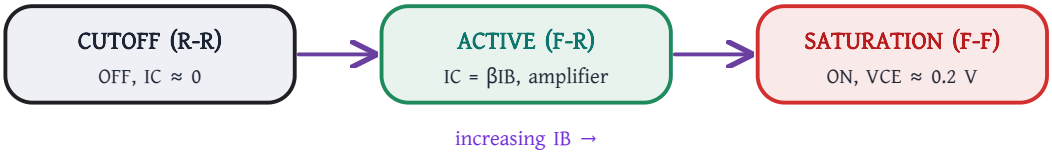
No runaway: $V_{CE} < V_{CC}/2$

Top 12 one-liners for MCQs

1. Doping E > C > B; width C > E > B
2. Base: thinnest and lightest doped
3. Active = J_E forward, J_C reverse
4. Saturation = both forward (switch ON)
5. CE: 180° phase shift, highest power gain
6. CC: highest Z_{in}, $A_v \approx 1$, buffer
7. CB: lowest Z_{in}, $A_i = \alpha < 1$
8. Most stable bias: voltage-divider
9. Least stable: fixed bias, $S = 1 + \beta$
10. npn faster than pnp (electron mobility)
11. Early effect = base-width modulation
12. Storage time t_s limits switching speed

Top traps: E and C are not interchangeable · in saturation $I_C < \beta I_B$ · CC current gain = $1 + \beta$, not β · $I_{CEO} \gg I_{CBO}$ · bypassing R_E keeps DC stability · β is not constant

All tricks together: npn = Not Pointing iN · α Almost 1, β Big · F-R = For Real gain, F-F fully ON, R-R really OFF · CB Below 1, CE gains Everything, CC Copies · R_E looks $(1 + \beta)$ times bigger from the base



one picture for all three regions

✓ Practice MCQs (exam pattern, 10 marks)

1. The most heavily doped region of a BJT is the:
(a) base (b) collector
(c) emitter (d) depletion layer

2. If $\alpha = 0.98$, β is:
(a) 49 (b) 98
(c) 0.02 (d) 50

3. In saturation, (E–B, C–B) are:
(a) forward, reverse (b) reverse, reverse
(c) reverse, forward (d) forward, forward

4. Highest input impedance, voltage gain ≈ 1:
(a) CB (b) CC
(c) CE (d) CE, bypassed R_E

5. I_{CEO} equals:
(a) I_{CBO} (b) βI_{CBO}
(c) $(1 + \beta)I_{CBO}$ (d) I_{CBO}/β

6. Fixed-bias S with $\beta = 99$:
(a) 1 (b) 99
(c) 50 (d) 100

7. r_e at $I_E = 2$ mA (room temp.):
(a) 13 Ω (b) 26 Ω
(c) 52 Ω (d) 1.3 Ω

8. $V_{CC} = 10$ V, $R_C = 2$ kΩ: $I_{C(sat)}$ =
(a) 2 mA (b) 20 mA
(c) 5 mA (d) 10 mA

9. The widest region of a BJT is the:
(a) emitter (b) base
(c) all equal (d) collector

10. Which delay is usually longest when a saturated BJT switches?
(a) delay t_d (b) storage t_s
(c) rise t_r (d) fall t_f

Answer key → 1-c · 2-a · 3-d · 4-b · 5-c · 6-d · 7-a · 8-c · 9-d · 10-b

Written question: Define stability factor S and derive it for voltage-divider bias. For $V_{CC} = 12$ V, $R_1 = 40$ kΩ, $R_2 = 10$ kΩ, $R_C = 2$ kΩ, $R_E = 1$ kΩ, $\beta = 100$, find the Q-point and S, and compare S with fixed bias. 10 marks

1. $S = \partial I_C/\partial I_{CBO}$; Thevenin: $V_{Th} = I_B R_{Th} + V_{BE} + (I_B + I_C)R_E \rightarrow \partial I_B/\partial I_C = -R_E/(R_{Th} + R_E) \rightarrow S = (1 + \beta)(1 + R_{Th}/R_E)/(1 + \beta + R_{Th}/R_E)$ 3
2. $\beta R_E = 100$ kΩ ≥ $10R_2 = 100$ kΩ ✓; $V_B = 12 \times 10/50 = 2.4$ V; $I_E = 1.7$ mA; $V_{CE} = 12 - 1.7 \times 3 =$ **6.9 V** 3
3. Exact check: $R_{Th} = 8$ kΩ, $I_B = 1.7/(8 + 101) = 15.6$ μA, $I_C = 1.56$ mA, $V_{CE} \approx 7.3$ V 1
4. $S = 101 \times 9/109 \approx$ **8.3** vs fixed bias 101: about 12 times more stable 3

Before the exam ✓

- ✓ Can draw npn/pnp symbols and layer doping
- ✓ Can convert $\alpha \leftrightarrow \beta$ and use I_{CEO}
- ✓ Know junction bias in all regions
- ✓ Can fill the CB/CE/CC table from memory
- ✓ Can solve bias circuits, draw the load line

Also practise (written):
① Explain thermal runaway
② S for collector-feedback
③ BJT inverter with R_B design
④ CE input/output curves